

PRASANNA SHRESTHA

Senior Software Engineer · Distributed Systems · Trust & Safety · Fraud & Abuse Prevention
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SUMMARY

Senior Software Engineer with 14+ years of experience building scalable backend services and full-stack applications, including 5 years at Intuit across data platform, abuse prevention, and Trust & Safety. Led a cross-team initiative to scale a fraud-risk service to 7,500 TPS and drove resilience work (rate limiting, disaster recovery). Strong in distributed systems, high-throughput service design, security and compliance, and applying modern tooling — including AI-assisted development — to ship faster. Proven owner of initiatives end to end, team lead, and mentor.

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6, Node.js, React, Redux), TypeScript, Python, GraphQL, C++, PHP

Backend & Distributed Systems: Spring Boot, Spring MVC, REST/SOAP, microservices, Kafka/RabbitMQ, rate limiting/throttling, caching, API gateway, high-availability & DR design, SOA

Cloud & Infra: AWS (EC2), Docker, Kubernetes/OpenShift, Jenkins CI/CD, Redis, Memcached, Apache Solr

Databases: MySQL, Oracle, MS SQL Server, PostgreSQL, MongoDB; Hibernate, JPA, Spring Data

Testing: Karate, WireMock, JUnit, Mockito, Jest, Playwright; contract, E2E, integration, TDD

Security & Trust/Safety: OAuth/SSO, secret rotation, CVE remediation, fraud/abuse prevention, phishing/spam detection, LLM-based risk scoring

AI-Assisted Development: Claude, custom AI commands/skills, MCP server integrations

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Intuit, CA July 2021 – July 2026

Trust & Safety / Product Abuse (STRA) · Abuse Prevention · Data Persistence Platform

- **Led the STRA scalability initiative** to scale a fraud-risk assessment service to 7,500 TPS, coordinating across upstream/downstream teams and adopting modern technologies to meet throughput and reliability targets.
- **Designed multi-layer rate limiting** (service, application, and gateway level, including per-client limits) that closed availability gaps surfaced through production incident RCAs.
- **Drove disaster recovery (DR) setup** for the service, strengthening resilience and business continuity.
- **Built the test-automation foundation** (Karate: E2E, component, and contract tests with WireMock) integrated into Jenkins CI/CD with automated PR-triggered runs; introduced AI-assisted development (Claude, MCP servers) to speed and improve test authoring.
- **Shipped a cost-optimization pre-check** in the phishing/spam detection pipeline that cut GenAI token usage and latency with no loss in detection accuracy; partnered with policy teams on new anti-spam policies.
- **Drove GED projects to production** end to end and migrated legacy identity infrastructure (Crowd) from colocation to managed SaaS — including enterprise SSO/OAuth and cloud DB setup.
- **Led post-incident RCAs, mentored engineers**, and designed data-persistence entities/schema and access-control models for consumer platform features. Recipient of the Engineering Excellence Award.

Software Engineer II — Expedia Inc., Chicago, IL Nov 2018 – Dec 2020

- Designed and built responsive UI features with React, Redux, and TypeScript in close partnership with UX; frequently guided UX design decisions.
- Built RESTful APIs and new services with Spring Boot, Node.js, and Kotlin, integrating third-party apps including a payment app in an admin tool.
- Raised quality with unit tests (JUnit, Jest, Enzyme) and partnered with PMs/UX to refine requirements and shape next-gen features.

Tech: React, TypeScript, MobX, Spring Boot, Node.js, Jest, Jenkins, Docker, AWS, PostgreSQL

Software Engineer II — Cisco Systems Inc., San Jose, CA Mar 2017 – Jul 2018

- Built interactive UIs (AngularJS, React) and server-side APIs (Node.js, JWT auth) for a Next-Generation Management Application; contributed to system and architecture design.
- Ran CI/CD and unit testing via Jenkins and Docker; set up VM test environments (ESXi, vCenter, vSphere); researched and shared new technologies with the team.

Tech: Angular.js, React, Node.js, D3.js, Mocha.js, Jenkins, Docker, OpenShift, Linux

Senior Software Developer & Team Lead — Javra Software, NepalJun 2015 – Apr 2016

- Led a 6-member team building a MEAN-stack web application; designed MongoDB schema, real-time notifications (Socket.io), Redis caching, and OAuth-secured REST APIs across the full SDLC (Agile).

Senior Software Engineer — Ace Express Travel Services, NepalFeb 2012 – May 2015

- Designed and deployed an online flight/hotel booking engine (Amadeus, TBO web services); built SOAP/REST APIs, automated invoicing/reporting (CRM), and SOA search (JMS/Node.js) — reducing manual paperwork.

Earlier — Web Developer roles, Nepal2009 – 2011

- Outlines R&D and Institute of Field & Research Expedition: built PHP/CodeIgniter web applications and a multimodal dictionary platform; co-authored a research paper (Oriental COCOSDA 2011).

EDUCATION

M.S. Computer Science — Maharishi University of Management, Fairfield, IA (2016–2018)

B.S. Information Management: Computer Science — Tribhuvan University / St. Xavier's College, Nepal (2004–2008)

CERTIFICATIONS & CONTINUING EDUCATION

- [Certification name] — [Issuer], [Year] (e.g., AWS Certified Solutions Architect – Associate)
- AI/ML coursework completed while at Intuit (2021–2026): [course/platform — e.g., DeepLearning.AI "Generative AI with LLMs", Coursera ML Specialization, Intuit internal GenAI training]
- [Add any additional certifications or courses]

PUBLICATION

Allwood, Dhakhwa, Pokharel, Regmi, Shrestha. "Multimodal Corpus Using Multimodal Dictionary in Lohorung," Oriental COCOSDA, 2011.

Note: [brackets] mark fields to confirm/fill — especially certifications, AI courses, and any metrics (traffic scale, % improvements, latency, cost saved).